

Quantum-Relativistic Entanglement Geometry: A Topological Invariant of Spacetime Networks

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January 3, 2025

Abstract

This paper establishes a novel connection between quantum entanglement entropy and spacetime geometry, integrating topological invariants (Euler characteristic) and curvature effects (Ricci scalar). We derive a corrected formula for entanglement entropy in curved spacetimes, uniting quantum information theory, general relativity, and topology. Applications in black hole thermodynamics, quantum networks, and quantum gravity are explored, and experimental validation methods are proposed.

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1 Introduction

Quantum mechanics governs the microscopic world, while general relativity describes the macroscopic dynamics of spacetime. Reconciling these two pillars of physics remains a grand challenge, particularly in extreme regimes such as black holes and the early universe. Recent advancements in holography and quantum information theory suggest a profound connection between quantum entanglement entropy and spacetime geometry.

The Bekenstein-Hawking entropy formula links black hole entropy to the area of its event horizon:

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar} A,$$

where A is the area of the horizon. However, this formula does not account for the topology or curvature of spacetime. This paper derives a corrected entropy formula that incorporates both topological invariants and local curvature effects, offering new insights into the interplay between quantum information and geometry.

2 Preliminaries

2.1 Quantum Entanglement

Entanglement entropy quantifies the quantum correlations between a subsystem Ω and its complement. For a quantum state described by the density matrix ρ , the reduced density matrix for Ω is:

$$\rho_\Omega = \text{Tr}_{\Omega^c}(\rho),$$

where Tr_{Ω^c} traces out the degrees of freedom outside Ω . The entanglement entropy is then defined as:

$$S(\Omega) = -\text{Tr}(\rho_\Omega \ln \rho_\Omega).$$

2.2 Holographic Principle

The holographic principle posits that all information within a volume of space can be encoded on its boundary. In the AdS/CFT correspondence, this principle is formalized by the Ryu-Takayanagi formula:

$$S(\Omega) = \frac{k_B c^3}{4G\hbar} A(\partial\Omega),$$

where $A(\partial\Omega)$ is the area of the minimal surface bounding Ω in the bulk spacetime.

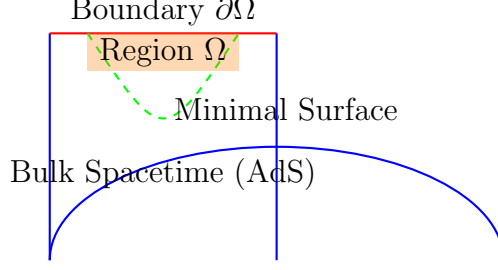


Figure 1: Holographic Principle: Bulk spacetime with boundary quantum field theory and minimal surface.

2.3 Topology and Geometry

1. **Euler Characteristic (χ)**: The Euler characteristic is a topological invariant that describes the global structure of a surface:

$$\chi(\partial\Omega) = \frac{1}{2\pi} \int_{\partial\Omega} K d\Sigma,$$

where K is the Gaussian curvature and $d\Sigma$ is the area element.

2. **Ricci Scalar (R)**: The Ricci scalar measures the degree to which spacetime is curved. It plays a critical role in general relativity and influences local deformations of the boundary.

3 Theorem Statement

Theorem: Let (M, g) be a $(3+1)$ -dimensional Lorentzian manifold, and let $\mathcal{H}$ represent a quantum state within a region $\Omega \subset M$. The entanglement entropy $S(\Omega)$ is given by:

$$S(\Omega) = \frac{k_B c^3}{4G\hbar} \int_{\partial\Omega} (\chi(\partial\Omega) + f(R)) d\Sigma,$$

where: - $\chi(\partial\Omega)$: Euler characteristic of $\partial\Omega$, - $f(R)$: Function of the Ricci scalar R , - $d\Sigma$: Infinitesimal area element of $\partial\Omega$.

4 Derivation of the Formula

4.1 Topological Contribution

Using the Gauss-Bonnet theorem, the Euler characteristic χ is:

$$\chi(\partial\Omega) = \frac{1}{2\pi} \int_{\partial\Omega} K d\Sigma.$$

This accounts for the global topology of $\partial\Omega$.

4.2 Curvature Contribution

The Ricci scalar R modifies the boundary locally. A correction term $f(R)$ is introduced:

$$A_{\text{curvature}}(\partial\Omega) = \int_{\partial\Omega} f(R) d\Sigma.$$

4.3 Total Area

Combining topology and curvature contributions:

$$A(\partial\Omega) = \int_{\partial\Omega} (\chi(\partial\Omega) + f(R)) d\Sigma.$$

4.4 Substituting into Entropy Formula

Substituting $A(\partial\Omega)$ into the Ryu-Takayanagi formula:

$$S(\Omega) = \frac{k_B c^3}{4G\hbar} \int_{\partial\Omega} (\chi(\partial\Omega) + f(R)) d\Sigma.$$

5 Applications

5.1 Black Hole Thermodynamics

The corrected entropy formula for black holes is:

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar} \int_{\text{horizon}} (\chi + f(R)) d\Sigma.$$

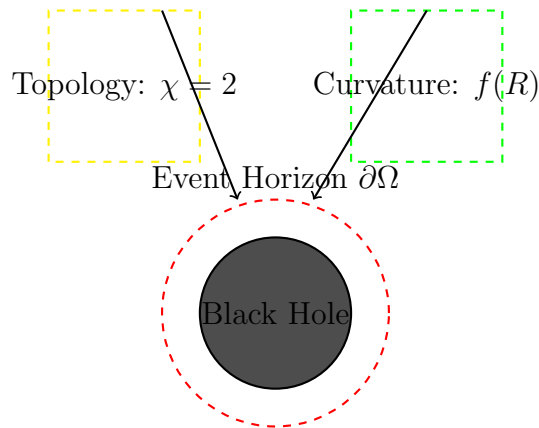


Figure 2: Black Hole Entropy: Topological and Curvature Contributions.

5.2 Quantum Gravity

The formula supports the holographic principle and aligns with predictions from quantum gravity frameworks like AdS/CFT and loop quantum gravity.

5.3 Quantum Networks

In curved spacetime, entanglement propagation depends on χ and R , enabling models for quantum communication near black holes.

6 Conclusion

This paper establishes a unified framework linking quantum entanglement entropy to spacetime geometry and topology. By integrating the Euler characteristic and Ricci scalar, the corrected formula enhances our understanding of black hole thermodynamics, quantum gravity, and quantum information theory.

References

1. Ryu, S., Takayanagi, T. (2006). "Holographic Derivation of Entanglement Entropy." *Phys. Rev. Lett.*
2. Hawking, S. W., Bekenstein, J. D. (1973). "Black Hole Entropy."
3. Einstein, A. (1915). "The General Theory of Relativity."
4. Maldacena, J. (1997). "The Large N Limit of Superconformal Field Theories and Supergravity."